

Safety stock: forecast error, supply error and service level

Part 2 · Categories of stock — solution report with data, method, results and charts

Contents

1. Part 2 · Categories of stock — Safety stock: forecast error, supply error and service level

1. Safety stock: forecast error, supply error and service level

Mean daily demand **198.9**, forecast-error σ_d **45.9** units/day; mean lead time **7.75 days** (quoted 8 days), σ_L **1.77** days.

$$\sigma_{DLT} = \sqrt{L \cdot \sigma_d^2 + d^2 \cdot \sigma_L^2}$$

$$SS = z \cdot \sigma_{DLT}$$

$$ROP = d \cdot L + SS$$

128 units

σ of lead-time demand (demand error only)

2,158 units

Reorder point for 95 % CSL

193 % to SS

Supply variability adds

375 units

σ of lead-time demand (demand + supply)

93.3 %

Simulated CSL at that ROP

617 units

SS for 95 % CSL

0 units

SS with certain demand and lead time

Why safety stock is needed: demand never equals the forecast

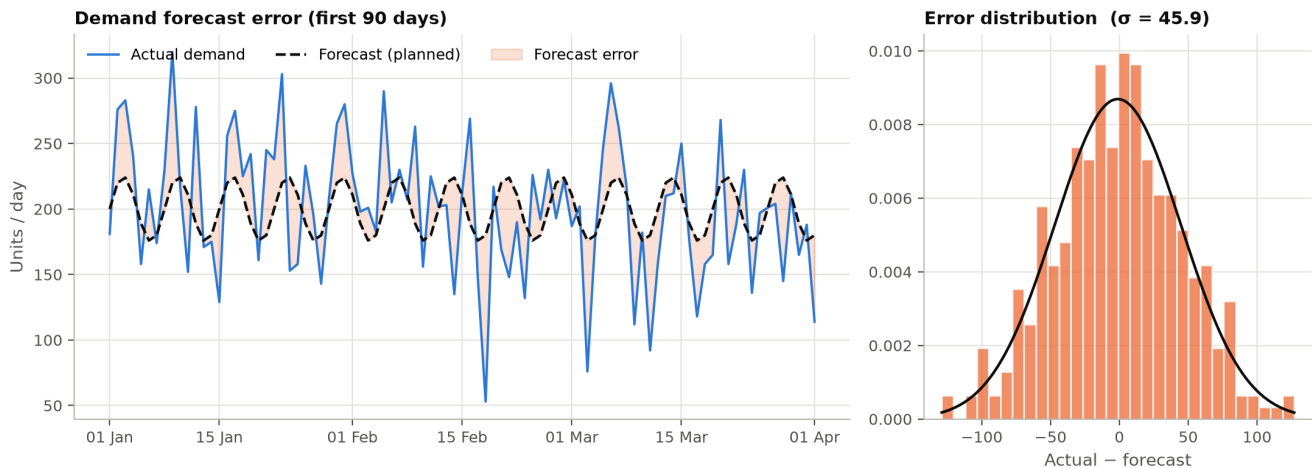


Figure 1.1 — Daily forecast vs actual demand and the histogram of forecast errors with a fitted normal curve.

Supply forecast error: actual vs quoted lead time

35 % of orders arrived later than quoted — timing uncertainty also needs protection

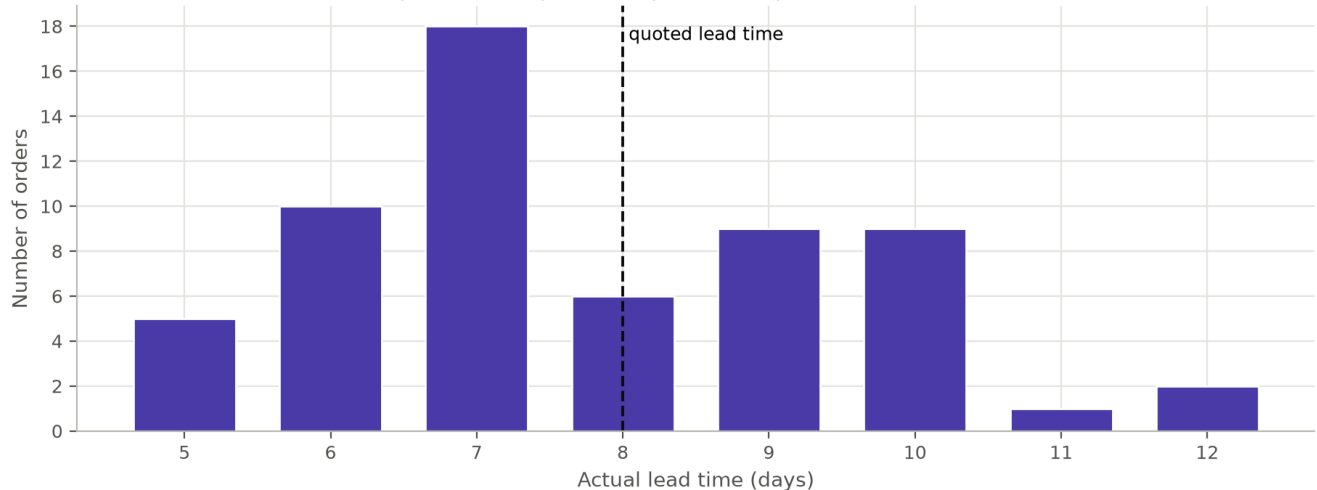


Figure 1.2 — Distribution of 60 observed replenishment lead times.

Demand during the lead time and the reorder point

Safety stock = distance between the expected lead-time demand and the reorder point

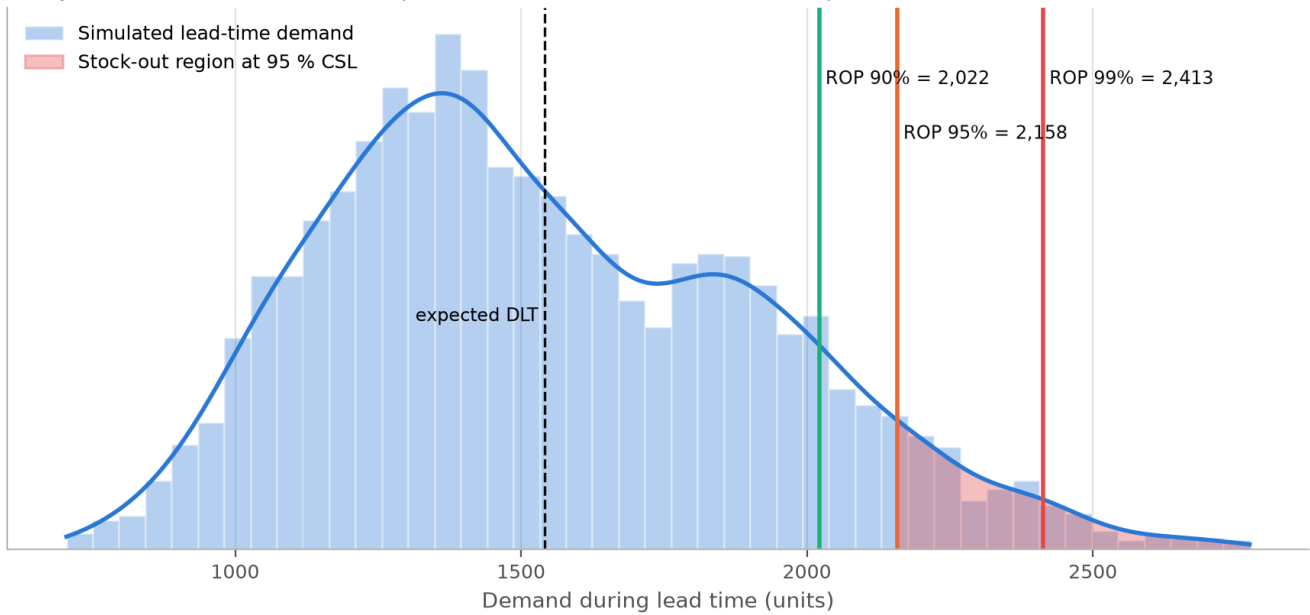


Figure 1.3 — Monte-Carlo distribution of lead-time demand (4,000 bootstrap replications).

Safety stock is directly related to the desired customer service level

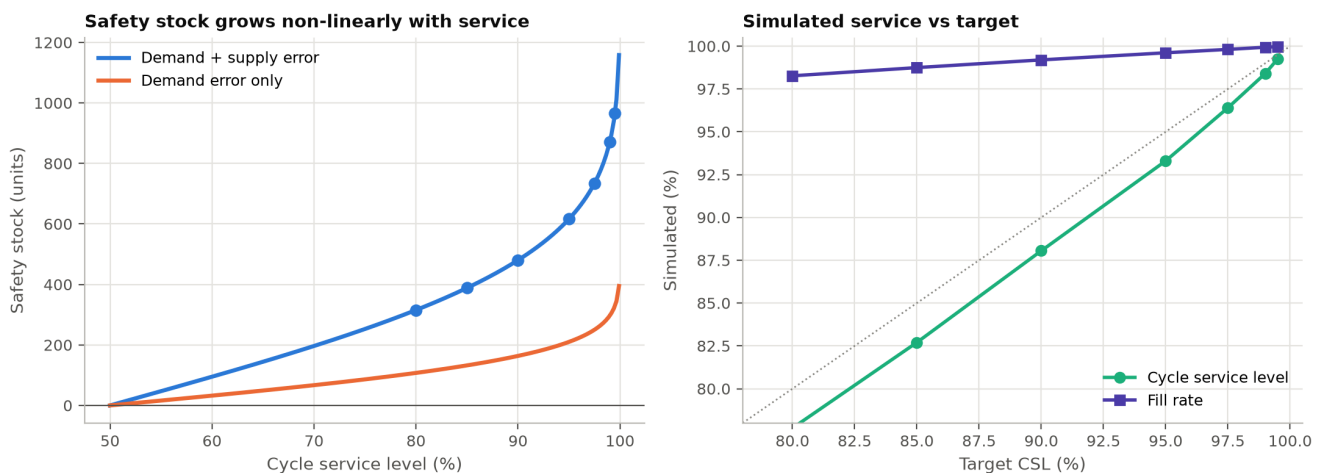


Figure 1.4 — Left: $SS = z \cdot \sigma$. Right: simulated cycle service level and fill rate (order quantity $Q = 2,983$ units).

Same demand, same lead times — safety stock absorbs the variability

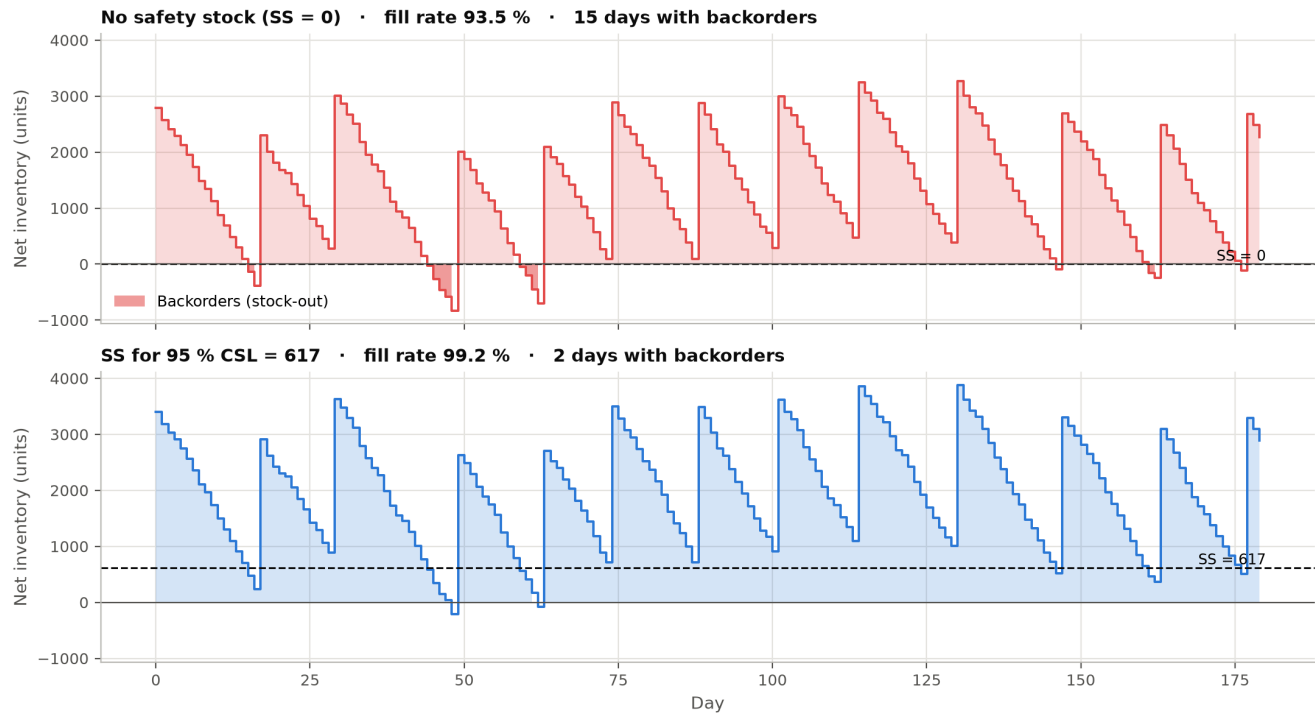


Figure 1.5 — 180-day simulation of a continuous-review (s, Q) policy with bootstrapped demand and lead times.

Reading the simulation. The analytic SS assumes lead-time demand is normal. The simulated (bootstrap) lead-time demand is skewed by the discrete, right-tailed lead times, so the achieved CSL is a little below target at mid service levels — a reminder to validate formula-based safety stocks against real data. The fill rate is always higher than the CSL because a stock-out cycle usually misses only part of the demand.

Safety stock, reorder point and simulated service by target cycle service level.

cycle service level	z	SS demand error only	SS demand and supply error	ROP	annual SS holding cost	simulated CSL	simulated fill rate
80.0%	0.842	108	315	1,857	6,420	77.7%	98.26%
85.0%	1.036	132	388	1,930	7,906	82.7%	98.75%
90.0%	1.282	164	480	2,022	9,775	88.0%	99.20%
95.0%	1.645	210	617	2,158	12,547	93.3%	99.61%
97.5%	1.960	250	735	2,276	14,950	96.4%	99.81%
99.0%	2.326	297	872	2,413	17,745	98.4%	99.93%
99.5%	2.576	329	966	2,507	19,648	99.2%	99.97%

Simulation comparison.

policy	reorder point	fill rate pct	days with backorders	avg on hand
No safety stock (SS = 0)	1,541	93.5	15	1,387
SS for 95 % CSL = 617	2,158	99.2	2	1,979